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Observe the Process Landscape

Dari output ps, terlihat sistem menjalankan berbagai jenis proses, mulai dari GUI (gnome-shell, PID 2608, %CPU 7.0%), system daemon (systemd, PID 1, %CPU 2.0%), kernel thread (migration/1-3 dengan PRI 139 dan NI -), dan shell (bash, PID 4606).

Setiap proses memiliki STAT yang mencerminkan kondisi kernel: Rsl untuk gnome-shell (running, multi-threaded), Ss untuk systemd (sleeping, session leader), dan S untuk bash (sleeping menunggu input). Output top menampilkan data yang sama secara real-time dengan tambahan kolom VIRT, RES, dan SHR.

Inspeksi /proc menunjukkan metadata kernel. Ketiga proses sama-sama berada di state S (sleeping), yang bukan berarti tidak aktif, melainkan sedang menunggu event seperti sinyal, I/O, atau input pengguna. /proc menyimpan metadata lengkap yang di-maintain langsung oleh kernel.

```
fakhruy@ubuntudesktop01:~$ ps -eo pid,ppid,user,stat,ni,pri,comm,%cpu,%mem --sort=-%cpu | head -20
```

PID	PPID	USER	STAT	NI	PRI	COMMAND	%CPU	%MEM
2608	2353	fakhruy	Rsl	0	19	gnome-shell	7.0	3.7
3134	2560	fakhruy	Sl	0	19	gvfsd-trash	4.4	0.4
2975	2717	fakhruy	Sl	0	19	ibus-extension-	2.5	0.3
1464	1	root	Ssl	0	19	snapt	2.3	0.5
1	0	root	Ss	0	19	systemd	2.0	0.1
3481	2353	fakhruy	Ssl	0	19	gnome-terminal-	0.9	0.7
2353	1	fakhruy	Ss	0	19	systemd	0.7	0.1
35	2	root	S	-	139	migration/3	0.6	0.0
23	2	root	S	-	139	migration/1	0.6	0.0
29	2	root	S	-	139	migration/2	0.6	0.0
3260	2608	fakhruy	Sl	0	19	gjs	0.6	0.8
388	1	root	S<s	-1	20	systemd-journal	0.4	0.2
2386	2353	fakhruy	Ss	0	19	dbus-daemon	0.4	0.0
1429	1	polkitd	Ssl	0	19	polkitd	0.4	0.1
40	2	root	I	0	19	kworker/u514:0-	0.3	0.0
1361	1	message+	Ss	0	19	dbus-daemon	0.3	0.0
2708	2353	fakhruy	Ssl	0	19	evolution-sourc	0.3	0.5
2802	2576	fakhruy	Sl	0	19	evolution-alarm	0.3	0.7
2792	2353	fakhruy	Sl	0	19	vmtoolsd	0.3	0.4

```
top - 13:11:13 up 8 min, 1 user, load average: 0.08, 0.20, 0.16
Tasks: 328 total, 1 running, 327 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.4 us, 0.2 sy, 0.0 ni, 99.4 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7893.8 total, 3700.7 free, 1403.5 used, 3098.9 buff/cache
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used, 6490.3 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2608	fakhruy	20	0	4450828	306252	140368	S	2.3	3.8	0:07.92	gnome-shell
4550	fakhruy	20	0	712760	62960	50592	S	1.7	0.8	0:00.96	gnome-terminal-
4571	fakhruy	20	0	23192	6040	3808	R	0.3	0.1	0:00.01	top
1	root	20	0	23604	15016	9868	S	0.0	0.2	0:02.17	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.01	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
11	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-events_highpri

```
fakhrusy@ubuntudesktop01:~$ cat /proc/2608/status
```

```
Name:   gnome-shell
Umask:  0002
State:  S (sleeping)
Tgid:   2608
Ngid:   0
Pid:    2608
PPid:   2353
TracerPid: 0
Uid:    1000    1000    1000    1000
Gid:    1000    1000    1000    1000
FDSize: 256
Groups: 4 24 27 30 46 100 114 1000
NSTgid: 2608
NSpid:  2608
NSpgid: 2608
NSSid:  2608
Kthreed: 0
VmPeak: 4864204 kB
VmSize: 4448700 kB
VmLck:  0 kB
VmPin:  0 kB
VmHWM:  310348 kB
VmRSS:  306048 kB
RssAnon:      165680 kB
RssFile:      124480 kB
RssShmem:      15888 kB
VmData:  294144 kB
VmStk:    132 kB
VmExe:     8 kB
VmLib:    216040 kB
VmPTE:    1348 kB
```

```
fakhrusy@ubuntudesktop01:~$ cat /proc/2608/stat
```

```
2608 (gnome-shell) R 2353 2608 2608 0 -1 4194304 80073 0 5 0 649 321 0 0 20 0 17 0 2821 4555468800 76467 18446744073709551615 105291547250688 105291547258577 140722736588048 0 0 0 0 16781312 17656 0 0 0 17 1 0 0 0 0 0 105291547265520 105291547267440 105291551158272 140722736593473 140722736593494 140722736593494 140722736594915 0
```

```
fakhrusy@ubuntudesktop01:~$ cat /proc/1/status
```

```
Name:   systemd
Umask:  0000
State:  S (sleeping)
Tgid:   1
Ngid:   0
Pid:    1
PPid:   0
TracerPid: 0
Uid:    0      0      0      0
Gid:    0      0      0      0
FDSize: 512
Groups:
NSTgid: 1
NSpid:  1
NSpgid: 1
NSSid:  1
Kthreed: 0
VmPeak: 23988 kB
VmSize: 23476 kB
VmLck:  0 kB
VmPin:  0 kB
VmHWM:  15012 kB
VmRSS:  15012 kB
RssAnon:      5144 kB
RssFile:      9868 kB
RssShmem:      0 kB
VmData:  4436 kB
VmStk:    132 kB
VmExe:     44 kB
VmLib:    12196 kB
```

```
fakhrusy@ubuntudesktop01:~$ cat /proc/1/stat
```

```
1 (systemd) S 0 1 1 0 -1 4194560 39465 321108 32 677 72 148 453 871 20 0 1 0 55 24039424 3667 18446744073709551615 1 1 0 0 0 0 671173123 4096 1260 0 0 0 17 1 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
```

```
fakhrusy@ubuntudesktop01:~$ cat /proc/1/cmdline
```

```
/sbin/init splash fakhrusy@ubuntudesktop01:~$
```

```
fakhrusy@ubuntu01:~$ cat /proc/4606/status
Name: bash
Umask: 0002
State: S (sleeping)
Tgid: 4606
Ngid: 0
Pid: 4606
Ppid: 4599
TracerPid: 0
Uid: 1000 1000 1000 1000
Gid: 1000 1000 1000 1000
FDSize: 256
Groups: 4 24 27 30 46 100 114 1000
NSTgid: 4606
NSpid: 4606
NSpgid: 4606
NSSid: 4606
Kthreadd: 0
VmPeak: 19940 kB
VmSize: 19940 kB
VmLck: 0 kB
VmPin: 0 kB
VmHWM: 5688 kB
VmRSS: 5688 kB

fakhrusy@ubuntu01:~$ cat /proc/4606/stat
4606 (bash) S 4599 4606 4606 34816 4699 4194304 1183 2085 0 0 1 0 0 2 20 0 1 0 54823 20418560 1365 18446744073709551615 11049316669
8496 110493167674157 140728977226576 0 0 0 65536 3686404 1266761467 1 0 0 17 2 0 0 0 0 0 110493167897008 110493167945648 1104934378
04544 140728977234319 140728977234324 140728977234324 140728977234324 140728977234324 140728977234324 140728977234324 140728977234324 0
fakhrusy@ubuntu01:~$ cat /proc/4606/cmdline
bashfakhrusy@ubuntu01:~$
```

Trace Parent-Child Relationships and Process Creation

Output `ps -p` memperlihatkan seluruh hierarki proses dimulai dari `systemd(1)` sebagai *root*. `sleep 300 &` menghasilkan data yang sangat informatif tentang job control. Output `jobs -l` menampilkan [1]+ 5798 Running sleep 300 &, dan `ps -o pid,ppid,pgid,sid,stat,cmd -p 5798` menunjukkan: PID = 5798, PPID = 5483 (bash), PGID = 5798, SID = 5483.

```

fakhrusy@ubuntudesktop01:~$ pstree -p
systemd(1)─ModemManager(1577)─{ModemManager}(1592)
                        │
                        ├──{ModemManager}(1593)
                        └──{ModemManager}(1596)
      ──NetworkManager(1509)─{NetworkManager}(1583)
                        │
                        ├──{NetworkManager}(1584)
                        └──{NetworkManager}(1585)
      ──VGAuthService(762)
      ──accounts-daemon(1466)─{accounts-daemon}(1496)
                        │
                        ├──{accounts-daemon}(1497)
                        └──{accounts-daemon}(1503)
      ──avahi-daemon(1360)─avahi-daemon(1508)
      ──colord(1940)─{colord}(1945)
                        │
                        ├──{colord}(1946)
                        └──{colord}(1948)
      ──cron(1474)
      ──cups-browsed(1718)─{cups-browsed}(1736)
                        │
                        ├──{cups-browsed}(1737)
                        └──{cups-browsed}(1738)
      ──cupsd(1680)
      ──dbus-daemon(1361)
      ──fwupd(4658)─{fwupd}(4659)
                        │
                        ├──{fwupd}(4660)
                        ├──{fwupd}(4661)
                        ├──{fwupd}(4662)
                        └──{fwupd}(4664)
      ──gdm3(1704)─gdm-session-wor(2341)─gdm-wayland-ses(2463)─gnome-session-b(2468)─{gnome-session-b}(2540)
                        │
                        ├──{gnome-session-b}(2541)
                        └──{gnome-session-b}(2542)
                        │
                        ├──{gdm-wayland-ses}(2465)
                        ├──{gdm-wayland-ses}(2466)
                        └──{gdm-wayland-ses}(2467)

```

```

fakhrusy@ubuntudesktop01:~$ ps -eo pid,ppid,pgid,sid,comm --forest
PID    PPID    PGID    SID    COMMAND
2       0       0       0      kthreadd
3       2       0       0      \_ pool_workqueue_release
4       2       0       0      \_ kworker/R-rcu_gp
5       2       0       0      \_ kworker/R-sync_wq
6       2       0       0      \_ kworker/R-kvfree_rcu_reclaim
7       2       0       0      \_ kworker/R-slub_flushwq
8       2       0       0      \_ kworker/R-netns
12      2       0       0      \_ kworker/u512:0-ipv6_addrconf
13      2       0       0      \_ kworker/R-mm_percpu_wq
14      2       0       0      \_ ksoftirqd/0
15      2       0       0      \_ rcu_preempt
16      2       0       0      \_ rcu_exp_par_gp_kthread_worker/1
17      2       0       0      \_ rcu_exp_gp_kthread_worker
18      2       0       0      \_ migration/0
19      2       0       0      \_ idle_inject/0
20      2       0       0      \_ cpuhp/0
21      2       0       0      \_ cpuhp/1
22      2       0       0      \_ idle_inject/1
23      2       0       0      \_ migration/1
24      2       0       0      \_ ksoftirqd/1
26      2       0       0      \_ kworker/1:0H-events_highpri
27      2       0       0      \_ cpuhp/2
28      2       0       0      \_ idle_inject/2
29      2       0       0      \_ migration/2
30      2       0       0      \_ ksoftirqd/2
32      2       0       0      \_ kworker/2:0H-events_highpri
33      2       0       0      \_ cpuhp/3
34      2       0       0      \_ idle_inject/3
35      2       0       0      \_ migration/3

```



```
fakhrusy@ubuntudesktop01:~$ sleep 300 & jobs -l
[1] 5798
[1]+  5798 Running                  sleep 300 &
fakhrusy@ubuntudesktop01:~$ ps -o pid,ppid,pgid,sid,stat,cmd -p $(pgrep sleep)
  PID   PPID   PGID   SID  STAT  CMD
  5798   5483   5798   5483  S     sleep 300
fakhrusy@ubuntudesktop01:~$ kill $(pgrep sleep)
fakhrusy@ubuntudesktop01:~$ ps -p $(pgrep sleep)
error: list of process IDs must follow -p

Usage:
ps [options]

Try 'ps --help <simple|list|output|threads|misc|all>'
  or 'ps --help <s|l|o|t|m|a>'
for additional help text.

For more details see ps(1).
[1]+  Terminated                  sleep 300
fakhrusy@ubuntudesktop01:~$
```

Inspect Open Resources

Perintah `lsuf | head -40` menampilkan daftar global resource yang sedang dibuka oleh semua proses. Sebagian besar baris awal menunjukkan pesan "Permission denied" karena `lsuf` dijalankan tanpa `sudo`, sehingga proses milik root (`systemd`, `kthreadd`, `pool_work`, `kworker`) tidak dapat dibaca symlink-nya di `/proc`.

Perintah `lsuf -i` dan `lsuf -i :22` menghasilkan output kosong karena tidak ada koneksi jaringan aktif yang dimiliki user.

```
fakhrusy@ubuntudesktop01:~$ lsuf -p 4606
COMMAND  PID    USER    FD    TYPE  DEVICE  SIZE/OFF  NODE  NAME
bash     4606  fakhrusy  cwd    DIR    8,2      4096  828581 /home/fakhrusy
bash     4606  fakhrusy  rtd    DIR    8,2      4096    2 /
bash     4606  fakhrusy  txt    REG    8,2     1446024  524865 /usr/bin/bash
bash     4606  fakhrusy  mem    REG    8,2     14596880  530338 /usr/lib/locale/locale-archive
bash     4606  fakhrusy  mem    REG    8,2     2125328  539846 /usr/lib/x86_64-linux-gnu/libc.so.6
bash     4606  fakhrusy  mem    REG    8,2     208328  540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
bash     4606  fakhrusy  mem    REG    8,2      27028  541603 /usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache
bash     4606  fakhrusy  mem    REG    8,2     236616  539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
bash     4606  fakhrusy   0u    CHR   136,0      0t0    3 /dev/pts/0
bash     4606  fakhrusy   1u    CHR   136,0      0t0    3 /dev/pts/0
bash     4606  fakhrusy   2u    CHR   136,0      0t0    3 /dev/pts/0
bash     4606  fakhrusy  255u   CHR   136,0      0t0    3 /dev/pts/0
```

```
fakhrusy@ubuntudesktop01:~$ lsof | head -40
COMMAND  PID  TID TASKCMD      USER  FD      TYPE          DEVICE  SIZE/OFF  NODE NAME
systemd  1                                root   cwd         unknown                               /proc/1/cwd (readlink: Permi
ssion denied)
systemd  1                                root   rtd         unknown                               /proc/1/root (readlink: Perm
ission denied)
systemd  1                                root   txt         unknown                               /proc/1/exe (readlink: Permi
ssion denied)
systemd  1                                root   NOFD                               /proc/1/fd (opendir: Permiss
ion denied)
kthreadd 2                                root   cwd         unknown                               /proc/2/cwd (readlink: Permi
ssion denied)
kthreadd 2                                root   rtd         unknown                               /proc/2/root (readlink: Perm
ission denied)
kthreadd 2                                root   txt         unknown                               /proc/2/exe (readlink: Permi
ssion denied)
kthreadd 2                                root   NOFD                               /proc/2/fd (opendir: Permiss
ion denied)
pool_work 3                                root   cwd         unknown                               /proc/3/cwd (readlink: Permi
ssion denied)
pool_work 3                                root   rtd         unknown                               /proc/3/root (readlink: Perm
ission denied)
pool_work 3                                root   txt         unknown                               /proc/3/exe (readlink: Permi
ssion denied)
pool_work 3                                root   NOFD                               /proc/3/fd (opendir: Permiss
ion denied)
kworker/R 4                                root   cwd         unknown                               /proc/4/cwd (readlink: Permi
ssion denied)
kworker/R 4                                root   rtd         unknown                               /proc/4/root (readlink: Perm
ission denied)
kworker/R 4                                root   txt         unknown                               /proc/4/exe (readlink: Permi
ssion denied)
```

```
fakhrusy@ubuntudesktop01:~$ lsof /tmp/testfile_lsof.txt
COMMAND  PID  USER  FD  TYPE  DEVICE  SIZE/OFF  NODE NAME
tail      5024 fakhrusy  3r  REG   8,2      0 1966160 /tmp/testfile_lsof.txt

fakhrusy@ubuntudesktop01:~$ lsof -i
fakhrusy@ubuntudesktop01:~$ lsof -i :22
fakhrusy@ubuntudesktop01:~$ lsof -i :63i
lsof: illegal option character: i
lsof 4.95.0
latest revision: https://github.com/lsof-org/lsof
latest FAQ: https://github.com/lsof-org/lsof/blob/master/00FAQ
latest (non-formatted) man page: https://github.com/lsof-org/lsof/blob/master/Lsof.8
usage: [-?abhKlnNoOPRtUvVX] [+|-c c] [+|-d s] [+D D] [+|-E] [+|-e s] [+|-f[gG]]
[-F [f]] [-g [s]] [-i [i]] [+|-L [l]] [+m [m]] [+|-M] [-o [o]] [-p s]
[+|-r [t]] [-s [p:s]] [-S [t]] [-T [t]] [-u s] [+|-w] [-x [fl]] [--] [names]
Use the ``-h'' option to get more help information.
fakhrusy@ubuntudesktop01:~$ lsof -i :63i
lsof: unacceptable port specification in: -i :63i
lsof 4.95.0
latest revision: https://github.com/lsof-org/lsof
latest FAQ: https://github.com/lsof-org/lsof/blob/master/00FAQ
latest (non-formatted) man page: https://github.com/lsof-org/lsof/blob/master/Lsof.8
usage: [-?abhKlnNoOPRtUvVX] [+|-c c] [+|-d s] [+D D] [+|-E] [+|-e s] [+|-f[gG]]
[-F [f]] [-g [s]] [-i [i]] [+|-L [l]] [+m [m]] [+|-M] [-o [o]] [-p s]
[+|-r [t]] [-s [p:s]] [-S [t]] [-T [t]] [-u s] [+|-w] [-x [fl]] [--] [names]
Use the ``-h'' option to get more help information.
fakhrusy@ubuntudesktop01:~$
```

Find Which Process Owns a Port or File

ss -tuln menampilkan semua socket TCP dan UDP yang sedang listen. Port aktif yang ditemukan adalah port 631 (TCP, pada 127.0.0.1:631 dan [::1]:631) milik layanan CUPS (printer daemon), serta port 53 (TCP/UDP, pada 127.0.0.53 dan 127.0.0.54) milik systemd-resolved (DNS resolver lokal). Perbandingan dengan netstat -tuln menunjukkan port yang sama, namun kolom PID/Program name seluruhnya berisi - dengan pesan eksplisit *"Not all processes could be identified"*. Perbedaan keduanya adalah ss merupakan tool modern pengganti netstat yang lebih cepat. Perintah fuser /usr/bin/bash langsung mengembalikan 4606e — angka adalah PID dan huruf e berarti executable (proses sedang menjalankan binary tersebut).

```
fakhrusy@ubuntudesktop01:~$ ss -ltnp | grep :631
LISTEN 0      4096      127.0.0.1:*        0.0.0.0:*
LISTEN 0      4096      [:::1]:*          [:::]:*
```

```
fakhrusy@ubuntudesktop01:~$ ss -tulnp
Netid      State      Recv-Q     Send-Q       Local Address:Port      Peer Address:Port      Process
udp        UNCONN     0           0            127.0.0.54:53           0.0.0.0:*
udp        UNCONN     0           0            127.0.0.53%lo:53        0.0.0.0:*
udp        UNCONN     0           0            0.0.0.0:57775          0.0.0.0:*
udp        UNCONN     0           0            0.0.0.0:5353           0.0.0.0:*
udp        UNCONN     0           0            [::]:42649             [::]:*
udp        UNCONN     0           0            [::]:5353              [::]:*
tcp        LISTEN     0           4096         127.0.0.1:631          0.0.0.0:*
tcp        LISTEN     0           4096         127.0.0.53%lo:53      0.0.0.0:*
tcp        LISTEN     0           4096         127.0.0.54:53         0.0.0.0:*
tcp        LISTEN     0           4096         [:::1]:631            [:::]:*
```

```
fakhrusy@ubuntudesktop01:~$ netstat -tulnp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 127.0.0.1:631          0.0.0.0:*              LISTEN      -
tcp        0      0 127.0.0.53:53         0.0.0.0:*              LISTEN      -
tcp        0      0 127.0.0.54:53         0.0.0.0:*              LISTEN      -
tcp6       0      0 :::1:631              :::*                   LISTEN      -
udp        0      0 127.0.0.54:53         0.0.0.0:*              -           -
udp        0      0 127.0.0.53:53         0.0.0.0:*              -           -
udp        0      0 0.0.0.0:57775         0.0.0.0:*              -           -
udp        0      0 0.0.0.0:5353          0.0.0.0:*              -           -
udp6       0      0 :::42649              :::*                   -           -
udp6       0      0 :::5353               :::*                   -           -
```

```
fakhrusy@ubuntu01:~$ fuser /usr/bin/bash
/usr/bin/bash: 4606e
fakhrusy@ubuntu01:~$ fuser -v /usr/bin/bash
USER      PID ACCESS COMMAND
/usr/bin/bash: fakhrusy 4606 ...e. bash
fakhrusy@ubuntu01:~$ fuser -vm /home
USER      PID ACCESS COMMAND
/home:     root      kernel swap  /swap.img
           root      kernel mount  /
           fakhrusy 2365 .rce. pipewire
           fakhrusy 2367 .rce. pipewire
           fakhrusy 2375 .rce. wireplumber
           fakhrusy 2376 .rce. pipewire-pulse
           fakhrusy 2380 .rce. gnome-keyring-d
           fakhrusy 2386 .rce. dbus-daemon
           fakhrusy 2428 .rce. xdg-document-po
           fakhrusy 2432 .rce. xdg-permission-
           fakhrusy 2463 .rce. gdm-wayland-ses
           fakhrusy 2468 .rce. gnome-session-b
           fakhrusy 2545 .rce. gcr-ssh-agent
           fakhrusy 2546 .rce. gnome-session-c
           fakhrusy 2560 .rce. gvfsd
           fakhrusy 2573 .rce. gvfsd-fuse
           fakhrusy 2576 Frce. gnome-session-b
           fakhrusy 2608 Frce. gnome-shell
           fakhrusy 2609 .rce. at-spi-bus-laun
           fakhrusy 2622 .rce. dbus-daemon
           fakhrusy 2679 .rce. at-spi2-registr
           fakhrusy 2690 .rce. gnome-shell-cal
           fakhrusy 2708 .rce. evolution-sourc
           fakhrusy 2710 .rce. gjs
```

Measure Per-Process CPU and I/O Activity

Proses `yes > /dev/null` (PID 5661) digunakan sebagai beban CPU sintetis. Hasil `pidstat -p 5661 1 10` merekam 10 interval per detik. Hasil `pidstat -r -p 5661 1 10` menampilkan data memori yang sepenuhnya statis selama 10 interval.

```
fakhrusy@ubuntu01:~$ pgrep yes
5661
fakhrusy@ubuntu01:~$ pidstat -p 5661 1 10
Linux 6.17.0-19-generic (ubuntu01)      04/29/2026      _x86_64_      (4 CPU)

01:55:02 PM  UID      PID      %usr %system %guest  %wait  %CPU   CPU   Command
01:55:03 PM  1000     5661    44.00  57.00   0.00   0.00  101.00    2  yes
01:55:04 PM  1000     5661    44.00  56.00   0.00   0.00  100.00    2  yes
01:55:05 PM  1000     5661    43.00  57.00   0.00   0.00  100.00    2  yes
01:55:06 PM  1000     5661    40.00  59.00   0.00   0.00   99.00    2  yes
01:55:07 PM  1000     5661    44.00  56.00   0.00   0.00  100.00    2  yes
01:55:08 PM  1000     5661    46.00  54.00   0.00   0.00  100.00    2  yes
01:55:09 PM  1000     5661    43.00  57.00   0.00   0.00  100.00    2  yes
01:55:10 PM  1000     5661    42.00  59.00   0.00   0.00  101.00    2  yes
01:55:11 PM  1000     5661    43.00  56.00   0.00   0.00   99.00    2  yes
01:55:12 PM  1000     5661    44.00  56.00   0.00   0.00  100.00    2  yes
Average:      1000     5661    43.30  56.70   0.00   0.00  100.00    -  yes

fakhrusy@ubuntu01:~$ pidstat -r -p 5661 1 10
Linux 6.17.0-19-generic (ubuntu01)      04/29/2026      _x86_64_      (4 CPU)

01:55:25 PM  UID      PID  minflt/s  majflt/s     VSZ     RSS   %MEM  Command
01:55:26 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:27 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:28 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:29 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:30 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:31 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:32 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:33 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:34 PM  1000     5661         0         0    16964    1900   0.02  yes
01:55:35 PM  1000     5661         0         0    16964    1900   0.02  yes
Average:      1000     5661         0         0    16964    1900   0.02  yes
```

Memory Pressure

Kondisi baseline sistem sebelum stress test menunjukkan: free -h melaporkan total RAM 7.7 GiB dengan used 1.5 GiB, free 2.5 GiB, buff/cache 4.0 GiB, dan available 6.3 GiB — swap 0B digunakan. MemFree adalah RAM yang benar-benar kosong, sedangkan MemAvailable adalah estimasi kernel tentang memori yang bisa segera digunakan tanpa swapping

```
fakhrusy@ubuntu01:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:          7.7Gi        1.5Gi        2.5Gi         42Mi        4.0Gi        6.3Gi
Swap:          4.0Gi           0B        4.0Gi

fakhrusy@ubuntu01:~$ vmstat 1 5
procs -----memory----- --swap--  -----io----- -system--  -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo    in    cs   us   sy   id   wa   st   gu
 0   0       0 2671476 66944 4165772    0    0    594    324   493    1    1    1  98    0    0    0
 0   0       0 2671476 66944 4165812    0    0      0      0    247   237    0    0  99    0    0    0
 0   0       0 2671300 66944 4165812    0    0      0      0   359   415    0    0  99    0    0    0
 0   0       0 2671236 66944 4165812    0    0      0      0   225   239    0    0 100    0    0    0
 0   0       0 2671236 66944 4165812    0    0      0      0   926  1299    1    0  99    0    0    0

fakhrusy@ubuntu01:~$ cat /proc/meminfo | head -40
MemTotal:        8083284 kB
MemFree:         2671108 kB
MemAvailable:    6556436 kB
Buffers:         66960 kB
Cached:          4000288 kB
SwapCached:      0 kB
Active:          2392952 kB
Inactive:        2272240 kB
Active(anon):    641780 kB
Inactive(anon):  0 kB
Active(file):    1751172 kB
Inactive(file):  2272240 kB
Unevictable:     16 kB
Mlocked:         16 kB
SwapTotal:       4194300 kB
SwapFree:        4194300 kB
Zswap:           0 kB
Zswapped:        0 kB
Dirty:           52 kB
Writeback:       0 kB
AnonPages:       597980 kB
Mapped:          357072 kB
Shmem:           43820 kB
KReclaimable:    165524 kB
Slab:            381840 kB
SReclaimable:    165524 kB
SUnreclaim:      216316 kB
KernelStack:     13104 kB
PageTables:      19488 kB
```

```
fakhrusy@ubuntudesktop01:~$ free -h & vmstat 1 10
[1] 6244
procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----
r b swpd free buff cache si so bi bo in cs us sy id wa st gu
4 0 0 2017484 68248 4217244 0 117 572 446 493 1 1 1 98 0 0 0
total used free shared buff/cache available
Mem: 7.7Gi 2.1Gi 1.9Gi 69Mi 4.1Gi 5.7Gi
Swap: 4.0Gi 0B 4.0Gi
1 0 0 2395096 68248 4217244 0 0 0 0 1339 424 24 1 75 0 0 0
1 0 0 2058764 68256 4217244 0 0 0 32840 1696 992 23 3 74 0 0 0
1 0 0 2057412 68256 4217244 0 0 0 0 1569 885 22 4 75 0 0 0
1 0 0 2453832 68256 4217244 0 0 0 1096 2017 1546 17 9 74 0 0 0
1 0 0 2056436 68256 4217244 0 0 0 0 1135 176 19 7 75 0 0 0
2 0 0 2350640 68256 4217244 0 0 0 0 1119 166 20 5 75 0 0 0
1 0 0 2155448 68268 4217244 0 0 0 1864 1141 177 20 6 75 0 0 0
1 0 0 2447920 68268 4217244 0 0 0 0 1216 299 15 11 75 0 0 0
1 0 0 2289976 68268 4217292 0 0 0 0 1725 1082 7 19 74 0 0 0
[1]+ Done free -h
```

Inspect a Process Memory Map with pmap and /proc/<PID>/maps

Output pmap 3463 pada proses sleep 300 (PID 3463) menampilkan virtual address space yang terbagi menjadi beberapa region dengan total 16968K. Perbandingan pmap \$\$ pada bash (PID 3336) dengan sleep mengungkap perbedaan kompleksitas. Bash memiliki mapping yang jauh lebih banyak: selain struktur yang sama (binary, heap, stack, libc, ld-linux), bash juga memuat libtinfo.so.6.4 (untuk terminal handling) dan gconv-modules.cache. Format /proc/<PID>/maps menampilkan informasi paling lengkap.

```
fakhrusy@ubuntudesktop01:~$ sleep 300 & echo "PID: $!"
```

```
[1] 3463
```

```
PID: 3463
```

```
fakhrusy@ubuntudesktop01:~$ pmap 3463
```

```
3463:  sleep 300
00005e2b5c3ea000      8K r---- sleep
00005e2b5c3ec000     16K r-x-- sleep
00005e2b5c3f0000      4K r---- sleep
00005e2b5c3f1000      4K r---- sleep
00005e2b5c3f2000      4K rw--- sleep
00005e2b722f3000    132K rw--- [ anon ]
00007d03df600000   14256K r---- locale-archive
00007d03e0400000    160K r---- libc.so.6
00007d03e0428000   1568K r-x-- libc.so.6
00007d03e05b0000   316K r---- libc.so.6
00007d03e05ff000    16K r---- libc.so.6
00007d03e0603000     8K rw--- libc.so.6
00007d03e0605000    52K rw--- [ anon ]
00007d03e0665000    12K rw--- [ anon ]
00007d03e0677000     4K r---- coreutils.mo
00007d03e0678000     8K rw--- [ anon ]
00007d03e067a000    16K r---- [ anon ]
00007d03e067e000     8K r---- [ anon ]
00007d03e0680000     8K r-x-- [ anon ]
00007d03e0682000     4K r---- ld-linux-x86-64.so.2
00007d03e0683000   172K r-x-- ld-linux-x86-64.so.2
00007d03e06ae000    40K r---- ld-linux-x86-64.so.2
00007d03e06b8000     8K r---- ld-linux-x86-64.so.2
00007d03e06ba000     8K rw--- ld-linux-x86-64.so.2
00007ffd5e150000   132K rw--- [ stack ]
fffffffff6000000     4K --x-- [ anon ]
total                16968K
```

```
fakhrusy@ubuntudesktop01:~$ cat /proc/3463/maps
```

```
5e2b5c3ea000-5e2b5c3ec000 r--p 00000000 08:02 525691 /usr/bin/sleep
5e2b5c3ec000-5e2b5c3f0000 r-xp 00002000 08:02 525691 /usr/bin/sleep
5e2b5c3f0000-5e2b5c3f1000 r--p 00006000 08:02 525691 /usr/bin/sleep
5e2b5c3f1000-5e2b5c3f2000 r--p 00007000 08:02 525691 /usr/bin/sleep
5e2b5c3f2000-5e2b5c3f3000 rw-p 00008000 08:02 525691 /usr/bin/sleep
5e2b722f3000-5e2b72314000 rw-p 00000000 00:00 0 [heap]
7d03df600000-7d03e03ec000 r--p 00000000 08:02 530338 /usr/lib/locale/locale-archive
7d03e0400000-7d03e0428000 r--p 00000000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
7d03e0428000-7d03e05b0000 r-xp 00028000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
7d03e05b0000-7d03e05ff000 r--p 001b0000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
7d03e05ff000-7d03e0603000 r--p 001fe000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
7d03e0603000-7d03e0605000 rw-p 00202000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
7d03e0605000-7d03e0612000 rw-p 00000000 00:00 0
7d03e0665000-7d03e0668000 rw-p 00000000 00:00 0
7d03e0677000-7d03e0678000 r--p 00000000 08:02 919592 /usr/share/locale-langpack/en/LC_MESSAGES/coreutils.mo
7d03e0678000-7d03e067a000 rw-p 00000000 00:00 0
7d03e067a000-7d03e067e000 r--p 00000000 00:00 0
7d03e067e000-7d03e0680000 r--p 00000000 00:00 0
7d03e0680000-7d03e0682000 r-xp 00000000 00:00 0
7d03e0682000-7d03e0683000 r--p 00000000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d03e0683000-7d03e06ae000 r-xp 00001000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d03e06ae000-7d03e06b8000 r--p 0002c000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d03e06b8000-7d03e06ba000 r--p 00036000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7d03e06ba000-7d03e06bc000 rw-p 00038000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffd5e150000-7ffd5e171000 rw-p 00000000 00:00 0 [stack]
fffffffff6000000-fffffffff6010000 --xp 00000000 00:00 0 [vsyscall]
```



```
fakhrusy@ubuntudesktop01:~$ pmap $$
3336:  bash
000065373f5f9000    192K r---- bash
000065373f629000    956K r-x-- bash
000065373f718000    212K r---- bash
000065373f74d000     16K r---- bash
000065373f751000     36K rw--- bash
000065373f75a000     44K rw--- [ anon ]
000065375e6ae000   1456K rw--- [ anon ]
000074207ac00000   14256K r---- locale-archive
000074207ba00000    160K r---- libc.so.6
000074207ba28000   1568K r-x-- libc.so.6
000074207bbb0000    316K r---- libc.so.6
000074207bbff000     16K r---- libc.so.6
000074207bc03000      8K rw--- libc.so.6
000074207bc05000    52K rw--- [ anon ]
000074207bc3f000    12K rw--- [ anon ]
000074207bc42000    56K r---- libtinfo.so.6.4
000074207bc50000    76K r-x-- libtinfo.so.6.4
000074207bc63000    56K r---- libtinfo.so.6.4
000074207bc71000    16K r---- libtinfo.so.6.4
000074207bc75000     4K rw--- libtinfo.so.6.4
000074207bc7f000    28K r--s- gconv-modules.cache
000074207bc86000     8K rw--- [ anon ]
000074207bc88000    16K r---- [ anon ]
000074207bc8c000     8K r---- [ anon ]
000074207bc8e000     8K r-x-- [ anon ]
000074207bc90000     4K r---- ld-linux-x86-64.so.2
000074207bc91000   172K r-x-- ld-linux-x86-64.so.2
000074207bc9c000    40K r---- ld-linux-x86-64.so.2
000074207bcc6000     8K r---- ld-linux-x86-64.so.2
000074207bcc8000     8K rw--- ld-linux-x86-64.so.2
```

```
fakhrusy@ubuntudesktop01:~$ cat /proc/$$/maps | head -40
65373f5f9000-65373f629000 r--p 00000000 08:02 524865 /usr/bin/bash
65373f629000-65373f718000 r-xp 00030000 08:02 524865 /usr/bin/bash
65373f718000-65373f74d000 r--p 0011f000 08:02 524865 /usr/bin/bash
65373f74d000-65373f751000 r--p 00154000 08:02 524865 /usr/bin/bash
65373f751000-65373f75a000 rw-p 00158000 08:02 524865 /usr/bin/bash
65373f75a000-65373f765000 rw-p 00000000 00:00 0
65375e6ae000-65375e81a000 rw-p 00000000 00:00 0 [heap]
74207ac00000-74207b9ec000 r--p 00000000 08:02 530338 /usr/lib/locale/locale-archive
74207ba00000-74207ba28000 r--p 00000000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207ba28000-74207bbb0000 r-xp 00028000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207bbb0000-74207bbff000 r--p 001b0000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207bbff000-74207bc03000 r--p 001fe000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207bc03000-74207bc05000 rw-p 00202000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207bc05000-74207bc12000 rw-p 00000000 00:00 0
74207bc3f000-74207bc42000 rw-p 00000000 00:00 0
74207bc42000-74207bc50000 r--p 00000000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc50000-74207bc63000 r-xp 0000e000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc63000-74207bc71000 r--p 00021000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc71000-74207bc75000 r--p 0002e000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc75000-74207bc76000 rw-p 00032000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc7f000-74207bc86000 r--s 00000000 08:02 541603 /usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache
74207bc86000-74207bc88000 rw-p 00000000 00:00 0
74207bc88000-74207bc8c000 r--p 00000000 00:00 0
74207bc8c000-74207bc8e000 r--p 00000000 00:00 0 [vvar]
74207bc8e000-74207bc90000 r-xp 00000000 00:00 0 [vvar_vclock]
74207bc90000-74207bc91000 r--p 00000000 00:00 0 [vdso]
74207bc91000-74207bc91000 r--p 00000000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74207bc91000-74207bc9c000 r-xp 00001000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74207bc9c000-74207bcc6000 r--p 0002c000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74207bcc6000-74207bcc8000 r--p 00036000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
```


Observe Shared Libraries and Dynamic Linking

Perintah ldd (List Dynamic Dependencies) digunakan untuk melihat shared libraries yang dibutuhkan oleh sebuah program. Pada /bin/ls, terdapat virtual dynamic shared object, library untuk security-enhanced linux, C standard library, dan Perl-Compatible Regular Expressions, dan Dynamic linker. Pada bash terdapat dependensi pada linux-vdso.so.1, libtinfo.so.6 (low-level terminal handling), libc.so.6, /lib64/ld-linux-x86-64.so.2.

```
fakhrusy@ubuntudesktop01:~$ ldd /bin/ls
linux-vdso.so.1 (0x0000772ea2715000)
libselinux.so.1 => /lib/x86_64-linux-gnu/libselinux.so.1 (0x0000772ea26ab000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x0000772ea2400000)
libpcre2-8.so.0 => /lib/x86_64-linux-gnu/libpcre2-8.so.0 (0x0000772ea2366000)
/lib64/ld-linux-x86-64.so.2 (0x0000772ea2717000)
fakhrusy@ubuntudesktop01:~$ ldd /bin/bash
linux-vdso.so.1 (0x000078ac4e588000)
libtinfo.so.6 => /lib/x86_64-linux-gnu/libtinfo.so.6 (0x000078ac4e3d0000)
libc.so.6 => /lib/x86_64-linux-gnu/libc.so.6 (0x000078ac4e000000)
/lib64/ld-linux-x86-64.so.2 (0x000078ac4e58a000)
```

```
fakhrusy@ubuntudesktop01:~$ cat /proc/3336/maps
65373f5f9000-65373f629000 r--p 00000000 08:02 524865 /usr/bin/bash
65373f629000-65373f718000 r-xp 00030000 08:02 524865 /usr/bin/bash
65373f718000-65373f74d000 r--p 0011f000 08:02 524865 /usr/bin/bash
65373f74d000-65373f751000 r--p 00154000 08:02 524865 /usr/bin/bash
65373f751000-65373f75a000 rw-p 00158000 08:02 524865 /usr/bin/bash
65373f75a000-65373f765000 rw-p 00000000 00:00 0
65375e6ae000-65375e81a000 rw-p 00000000 00:00 0
74207ab7f000-74207ac00000 rw-p 00000000 00:00 0
74207ac00000-74207b9ec000 r--p 00000000 08:02 530338 /usr/lib/locale/locale-archive
74207ba00000-74207ba28000 r--p 00000000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207ba28000-74207bbb0000 r-xp 00028000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207bbb0000-74207bbff000 r--p 001b0000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207bbff000-74207bc03000 r--p 001fe000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207bc03000-74207bc05000 rw-p 00202000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
74207bc05000-74207bc12000 rw-p 00000000 00:00 0
74207bc12000-74207bc42000 rw-p 00000000 00:00 0
74207bc42000-74207bc50000 r--p 00000000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc50000-74207bc63000 r-xp 0000e000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc63000-74207bc71000 r--p 00021000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc71000-74207bc75000 r--p 0002e000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc75000-74207bc76000 rw-p 00032000 08:02 540459 /usr/lib/x86_64-linux-gnu/libtinfo.so.6.4
74207bc76000-74207bc86000 r--s 00000000 08:02 541603 /usr/lib/x86_64-linux-gnu/gconv/gconv-modules.cache
74207bc86000-74207bc88000 rw-p 00000000 00:00 0
74207bc88000-74207bc8c000 r--p 00000000 00:00 0
74207bc8c000-74207bc8e000 r--p 00000000 00:00 0
74207bc8e000-74207bc90000 r-xp 00000000 00:00 0
74207bc90000-74207bc91000 r--p 00000000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74207bc91000-74207bc9c000 r-xp 00001000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74207bc9c000-74207bcc6000 r--p 0002c000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
74207bcc6000-74207bcc8000 r--p 00036000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
```

Examine Executable File Structure

Header elf menggunakan format 64-bit dengan format data little endian. Tercatat sebagai DYN (Position-Independent Executable File) yang berarti kode program dapat dimuat ke memory address mana pun saat runtime. Section headers menampilkan pemisahan jenis data ke dalam blok yang spesifik, seperti .text, .rodata, .data, .bss. Program Headers memegang peran eksekusi dengan mengelompokkan sections berizin akses serupa (seperti Read-Execute atau Read-Write) ke dalam Segments agar siap diproses oleh kernel dan dynamic loader saat runtime. shared libraries (seperti libselinux.so.1 dan libc.so.6) yang dibutuhkan oleh suatu program dideklarasikan secara eksplisit di dalam struktur biner.

```
Fakhrusy@ubuntudesktop01:~$ readelf -h /bin/ls
ELF Header:
  Magic:   7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 00
  Class:                               ELF64
  Data:                               2's complement, little endian
  Version:                             1 (current)
  OS/ABI:                              UNIX - System V
  ABI Version:                         0
  Type:                                DYN (Position-Independent Executable file)
  Machine:                             Advanced Micro Devices X86-64
  Version:                             0x1
  Entry point address:                 0x6d30
  Start of program headers:            64 (bytes into file)
  Start of section headers:           140328 (bytes into file)
  Flags:                               0x0
  Size of this header:                 64 (bytes)
  Size of program headers:             56 (bytes)
  Number of program headers:           13
  Size of section headers:            64 (bytes)
  Number of section headers:           31
  Section header string table index: 30
```

```
fakhrusy@ubuntudesktop01:~$ readelf -S /bin/ls
There are 31 section headers, starting at offset 0x22428:
```

Section Headers:

[Nr]	Name	Type	Address	Offset
	Size	EntSize	Flags Link Info Align	
[0]		NULL	0000000000000000	00000000
	0000000000000000	0000000000000000	0 0 0	
[1]	.interp	PROGBITS	0000000000000318	00000318
	000000000000001c	0000000000000000	A 0 0 1	
[2]	.note.gnu.pr[...]	NOTE	0000000000000338	00000338
	0000000000000030	0000000000000000	A 0 0 8	
[3]	.note.gnu.bu[...]	NOTE	0000000000000368	00000368
	0000000000000024	0000000000000000	A 0 0 4	
[4]	.note.ABI-tag	NOTE	000000000000038c	0000038c
	0000000000000020	0000000000000000	A 0 0 4	
[5]	.gnu.hash	GNU_HASH	00000000000003b0	000003b0
	0000000000000050	0000000000000000	A 6 0 8	
[6]	.dynsym	DYNSYM	0000000000000400	00000400
	0000000000000c00	0000000000000018	A 7 1 8	
[7]	.dynstr	STRTAB	0000000000001000	00001000
	0000000000000614	0000000000000000	A 0 0 1	
[8]	.gnu.version	VERSYM	0000000000001614	00001614
	0000000000000100	0000000000000002	A 6 0 2	
[9]	.gnu.version_r	VERNEED	0000000000001718	00001718
	00000000000000f0	0000000000000000	A 7 2 8	
[10]	.rela.dyn	RELA	0000000000001808	00001808
	0000000000001530	0000000000000018	A 6 0 8	
[11]	.rela.plt	RELA	0000000000002d38	00002d38
	00000000000009c0	0000000000000018	AI 6 25 8	
[12]	.init	PROGBITS	0000000000004000	00004000
	000000000000001b	0000000000000000	AX 0 0 4	
[13]	.plt	PROGBITS	0000000000004020	00004020
	0000000000000690	0000000000000010	AX 0 0 16	
[14]	.plt.got	PROGBITS	00000000000046b0	000046b0

```
fakhrusy@ubuntudesktop01:~$ readelf -l /bin/ls
```

```
Elf file type is DYN (Position-Independent Executable file)
Entry point 0x6d30
There are 13 program headers, starting at offset 64
```

Program Headers:

Type	Offset	VirtAddr	PhysAddr
	FileSiz	MemSiz	Flags Align
PHDR	0x0000000000000040	0x0000000000000040	0x0000000000000040
	0x00000000000002d8	0x00000000000002d8	R 0x8
INTERP	0x0000000000000318	0x0000000000000318	0x0000000000000318
	0x000000000000001c	0x000000000000001c	R 0x1
[Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]			
LOAD	0x0000000000000000	0x0000000000000000	0x0000000000000000
	0x0000000000000368	0x0000000000000368	R 0x1000
LOAD	0x0000000000000400	0x0000000000000400	0x0000000000000400
	0x00000000000014f41	0x00000000000014f41	R E 0x1000
LOAD	0x00000000000019000	0x00000000000019000	0x00000000000019000
	0x000000000000071b0	0x000000000000071b0	R 0x1000
LOAD	0x00000000000020f30	0x00000000000021f30	0x00000000000021f30
	0x0000000000001348	0x00000000000025e8	RW 0x1000
DYNAMIC	0x00000000000021a38	0x00000000000022a38	0x00000000000022a38
	0x0000000000000200	0x0000000000000200	RW 0x8
NOTE	0x00000000000000338	0x00000000000000338	0x00000000000000338
	0x0000000000000030	0x0000000000000030	R 0x8
NOTE	0x00000000000000368	0x00000000000000368	0x00000000000000368
	0x0000000000000044	0x0000000000000044	R 0x4
GNU_PROPERTY	0x00000000000000338	0x00000000000000338	0x00000000000000338
	0x0000000000000030	0x0000000000000030	R 0x8
GNU_EH_FRAME	0x0000000000001e170	0x0000000000001e170	0x0000000000001e170
	0x000000000000005ec	0x000000000000005ec	R 0x4
GNU_STACK	0x0000000000000000	0x0000000000000000	0x0000000000000000
	0x0000000000000000	0x0000000000000000	RW 0x10
GNU_RELRO	0x00000000000020f30	0x00000000000021f30	0x00000000000021f30
	0x00000000000010d0	0x00000000000010d0	R 0x1

```
fakhrusy@ubuntu01:~$ readelf -d /bin/ls

Dynamic section at offset 0x21a38 contains 28 entries:
Tag              Type              Name/Value
0x0000000000000001 (NEEDED)          Shared library: [libselinux.so.1]
0x0000000000000001 (NEEDED)          Shared library: [libc.so.6]
0x000000000000000c (INIT)            0x4000
0x000000000000000d (FINI)            0x18f34
0x0000000000000019 (INIT_ARRAY)       0x21f30
0x000000000000001b (INIT_ARRAYSZ)      8 (bytes)
0x000000000000001a (FINI_ARRAY)       0x21f30
0x000000000000001c (FINI_ARRAYSZ)      8 (bytes)
0x000000006ffffef5 (GNU_HASH)         0x3b0
0x0000000000000005 (STRTAB)           0x1000
0x0000000000000006 (SYMTAB)           0x400
0x000000000000000a (STRSZ)            1556 (bytes)
0x000000000000000b (SYMMENT)          24 (bytes)
0x0000000000000015 (DEBUG)            0x0
0x0000000000000003 (PLTGOT)           0x22c38
0x0000000000000002 (PLTRELSZ)         2496 (bytes)
0x0000000000000014 (PLTREL)           RELA
0x0000000000000017 (JMPREL)           0x2d38
0x0000000000000007 (RELA)             0x1808
0x0000000000000008 (RELASZ)           5424 (bytes)
0x0000000000000009 (RELAENT)          24 (bytes)
0x000000000000001e (FLAGS)            BIND_NOW
0x000000006fffffbb (FLAGS_1)          Flags: NOW PIE
0x000000006ffffffe (VERNEED)          0x1718
0x000000006fffffff (VERNEEDNUM)       2
0x000000006fffff00 (VERSYM)           0x1614
0x000000006fffff09 (RELACOUNT)        212
0x0000000000000000 (NULL)             0x0
```

Disassemble Machine Code

Perintah `objdump -h` menampilkan daftar section di dalam berkas biner. Proses disassembly membuat kita dapat melihat instruksi assembly dan byte heksadesimal yang diproses CPU. `objdump -x` memperlihatkan seluruh header berkas.

```
fakhrusy@ubuntu01:~$ objdump -h /bin/ls

/bin/ls:      file format elf64-x86-64

Sections:
Idx Name          Size      VMA               LMA               File off  Algn
 0 .interp         0000001c  0000000000000318  0000000000000318  00000318  2**0
CONTENTS, ALLOC, LOAD, READONLY, DATA
 1 .note.gnu.property 00000030  0000000000000338  0000000000000338  00000338  2**3
CONTENTS, ALLOC, LOAD, READONLY, DATA
 2 .note.gnu.build-id 00000024  0000000000000368  0000000000000368  00000368  2**2
CONTENTS, ALLOC, LOAD, READONLY, DATA
 3 .note.ABI-tag    00000020  000000000000038c  000000000000038c  0000038c  2**2
CONTENTS, ALLOC, LOAD, READONLY, DATA
 4 .gnu.hash        00000050  00000000000003b0  00000000000003b0  000003b0  2**3
CONTENTS, ALLOC, LOAD, READONLY, DATA
 5 .dynsym          00000c00  0000000000000400  0000000000000400  00000400  2**3
CONTENTS, ALLOC, LOAD, READONLY, DATA
 6 .dynstr          00000614  0000000000001000  0000000000001000  00001000  2**0
CONTENTS, ALLOC, LOAD, READONLY, DATA
 7 .gnu.version     00000100  0000000000001614  0000000000001614  00001614  2**1
CONTENTS, ALLOC, LOAD, READONLY, DATA
 8 .gnu.version_r   000000f0  0000000000001718  0000000000001718  00001718  2**3
CONTENTS, ALLOC, LOAD, READONLY, DATA
 9 .rela.dyn        00001530  0000000000001808  0000000000001808  00001808  2**3
CONTENTS, ALLOC, LOAD, READONLY, DATA
10 .rela.plt        000009c0  0000000000002d38  0000000000002d38  00002d38  2**3
CONTENTS, ALLOC, LOAD, READONLY, DATA
11 .init            0000001b  0000000000004000  0000000000004000  00004000  2**2
CONTENTS, ALLOC, LOAD, READONLY, CODE
12 .plt             00000690  0000000000004020  0000000000004020  00004020  2**4
CONTENTS, ALLOC, LOAD, READONLY, CODE
```

```
/bin/ls: file format elf64-x86-64
```

Disassembly of section .init:

```
0000000000004000 <.init>:
4000: f3 0f 1e fa      endbr64
4004: 48 83 ec 08      sub    $0x8,%rsp
4008: 48 8b 05 b9 ef 01 00 mov    0x1efb9(%rip),%rax      # 22fc8 <__gmon_start__@Base>
400f: 48 85 c0          test   %rax,%rax
4012: 74 02            je     4016 <free@plt-0x69a>
4014: ff d0            call   *%rax
4016: 48 83 c4 08      add    $0x8,%rsp
401a: c3              ret
```

Disassembly of section .plt:

```
0000000000004020 <.plt>:
4020: ff 35 1a ec 01 00 push   0x1ec1a(%rip)      # 22c40 <_obstack_memory_used@@Base+0x14d30>
4026: ff 25 1c ec 01 00 jmp     *0x1ec1c(%rip)      # 22c48 <_obstack_memory_used@@Base+0x14d38>
402c: 0f 1f 40 00      nopl   0x0(%rax)
4030: f3 0f 1e fa      endbr64
4034: 68 00 00 00 00 00 push   $0x0
4039: e9 e2 ff ff ff   jmp     4020 <free@plt-0x690>
403e: 66 90            xchg    %ax,%ax
4040: f3 0f 1e fa      endbr64
4044: 68 01 00 00 00 00 push   $0x1
4049: e9 d2 ff ff ff   jmp     4020 <free@plt-0x690>
404e: 66 90            xchg    %ax,%ax
4050: f3 0f 1e fa      endbr64
```

```
/bin/ls: file format elf64-x86-64
```

```
/bin/ls
```

```
architecture: i386:x86-64, flags 0x00000150:
```

```
HAS_SYMS, DYNAMIC, D_PAGED
```

```
start address 0x00000000000006d30
```

Program Header:

```
PHDR off 0x0000000000000040 vaddr 0x0000000000000040 paddr 0x0000000000000040 align 2**3
      filesz 0x00000000000002d8 memsz 0x00000000000002d8 flags r--
INTERP off 0x0000000000000318 vaddr 0x0000000000000318 paddr 0x0000000000000318 align 2**0
      filesz 0x000000000000001c memsz 0x000000000000001c flags r--
LOAD off 0x0000000000000000 vaddr 0x0000000000000000 paddr 0x0000000000000000 align 2**12
      filesz 0x000000000000036f8 memsz 0x000000000000036f8 flags r--
LOAD off 0x0000000000000400 vaddr 0x0000000000000400 paddr 0x0000000000000400 align 2**12
      filesz 0x00000000000014f41 memsz 0x00000000000014f41 flags r-x
LOAD off 0x0000000000001900 vaddr 0x0000000000001900 paddr 0x0000000000001900 align 2**12
      filesz 0x000000000000071b0 memsz 0x000000000000071b0 flags r--
LOAD off 0x00000000000020f30 vaddr 0x00000000000021f30 paddr 0x00000000000021f30 align 2**12
      filesz 0x0000000000001348 memsz 0x00000000000025e8 flags rw-
DYNAMIC off 0x00000000000021a38 vaddr 0x00000000000022a38 paddr 0x00000000000022a38 align 2**3
      filesz 0x0000000000000200 memsz 0x0000000000000200 flags rw-
NOTE off 0x0000000000000338 vaddr 0x0000000000000338 paddr 0x0000000000000338 align 2**3
      filesz 0x0000000000000030 memsz 0x0000000000000030 flags r--
NOTE off 0x0000000000000368 vaddr 0x0000000000000368 paddr 0x0000000000000368 align 2**2
      filesz 0x0000000000000044 memsz 0x0000000000000044 flags r--
0x6474e553 off 0x0000000000000338 vaddr 0x0000000000000338 paddr 0x0000000000000338 align 2**3
      filesz 0x0000000000000030 memsz 0x0000000000000030 flags r--
EH_FRAME off 0x0000000000001e170 vaddr 0x0000000000001e170 paddr 0x0000000000001e170 align 2**2
      filesz 0x000000000000005ec memsz 0x000000000000005ec flags r--
STACK off 0x0000000000000000 vaddr 0x0000000000000000 paddr 0x0000000000000000 align 2**4
```

Relate ELF Segments to Runtime Memory Layout

Pada saat menjalankan `readelf -l` pada `/bin/sleep`, terdapat program headers dengan segment bertipe load. Setiap segment memiliki permission yang spesifik, ada yang read, read-execute, read-write. Alamat awal pada memory map menunjukkan file `/usr/bin/sleep` yang di-map ke dalam beberapa region berbeda sesuai dengan permission aslinya.

```
Section to Segment mapping:
Segment Sections...
00
01      .interp
02      .interp .note.gnu.property .note.gnu.build-id .note.ABI-tag .gnu.hash .dynsym .dynstr .gnu.version .gnu.version_r .rela.d
yn .rela.plt
03      .init .plt .plt.got .plt.sec .text .fini
04      .rodata .eh_frame_hdr .eh_frame
05      .init_array .fini_array .data.rel.ro .dynamic .got .data .bss
06      .dynamic
07      .note.gnu.property
08      .note.gnu.build-id .note.ABI-tag
09      .note.gnu.property
10      .eh_frame_hdr
11
12      .init_array .fini_array .data.rel.ro .dynamic .got
```

```
fakhrusy@buntudesktop01:~$ readelf -l /bin/sleep
```

Elf file type is DYN (Position-Independent Executable file)

Entry point 0x2b80

There are 13 program headers, starting at offset 64

Program Headers:

Type	Offset FileSiz	VirtAddr MemSiz	PhysAddr Flags Align
PHDR	0x0000000000000040	0x0000000000000040	0x0000000000000040 R 0x8
INTERP	0x00000000000002d8	0x00000000000002d8	0x0000000000000318 R 0x1
[Requesting program interpreter: /lib64/ld-linux-x86-64.so.2]			
LOAD	0x000000000000012c0	0x000000000000012c0	0x0000000000000000 R 0x1000
LOAD	0x00000000000002000	0x00000000000002000	0x00000000000002000 R E 0x1000
LOAD	0x00000000000006000	0x00000000000006000	0x00000000000006000 R 0x1000
LOAD	0x00000000000007bd0	0x00000000000007bd0	0x00000000000007bd0 RW 0x1000
DYNAMIC	0x00000000000007c40	0x00000000000007c40	0x00000000000007c40 RW 0x8
NOTE	0x0000000000000338	0x0000000000000338	0x0000000000000338 R 0x8
NOTE	0x0000000000000368	0x0000000000000368	0x0000000000000368 R 0x4
GNU_PROPERTY	0x0000000000000338	0x0000000000000338	0x0000000000000338 R 0x8
GNU_EH_FRAME	0x00000000000006a80	0x00000000000006a80	0x00000000000006a80 R 0x4


```
fakhrusy@ubuntudesktop01:~$ sleep 300 &
[1] 3832
fakhrusy@ubuntudesktop01:~$ cat /proc/3832/maps
56229d202000-56229d204000 r--p 00000000 08:02 525691 /usr/bin/sleep
56229d204000-56229d208000 r-xp 00002000 08:02 525691 /usr/bin/sleep
56229d208000-56229d209000 r--p 00006000 08:02 525691 /usr/bin/sleep
56229d209000-56229d20a000 r--p 00007000 08:02 525691 /usr/bin/sleep
56229d20a000-56229d20b000 rw-p 00008000 08:02 525691 /usr/bin/sleep
5622acc0d000-5622acc2e000 rw-p 00000000 00:00 0 [heap]
71d690800000-71d6915ec000 r--p 00000000 08:02 530338 /usr/lib/locale/locale-archive
71d691600000-71d691628000 r--p 00000000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
71d691628000-71d6917b0000 r-xp 00028000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
71d6917b0000-71d6917ff000 r--p 001b0000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
71d6917ff000-71d691803000 r--p 001fe000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
71d691803000-71d691805000 rw-p 00202000 08:02 539846 /usr/lib/x86_64-linux-gnu/libc.so.6
71d691805000-71d691812000 rw-p 00000000 00:00 0
71d6918ca000-71d6918cd000 rw-p 00000000 00:00 0
71d6918dc000-71d6918dd000 r--p 00000000 08:02 919592 /usr/share/locale-langpack/en/LC_MESSAGES/coreutils.mo
71d6918dd000-71d6918df000 rw-p 00000000 00:00 0
71d6918df000-71d6918e3000 r--p 00000000 00:00 0 [vvar]
71d6918e3000-71d6918e5000 r--p 00000000 00:00 0 [vvar_vclock]
71d6918e5000-71d6918e7000 r-xp 00000000 00:00 0 [vdso]
71d6918e7000-71d6918e8000 r--p 00000000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
71d6918e8000-71d691913000 r-xp 00001000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
71d691913000-71d69191d000 r--p 0002c000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
71d69191d000-71d69191f000 r--p 00036000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
71d69191f000-71d691921000 rw-p 00038000 08:02 539649 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7fffdf8b0000-7fffdf8b0000 rw-p 00000000 00:00 0 [stack]
ffffffffff600000-ffffffffff601000 --xp 00000000 00:00 0 [vsyscall]
```

Observe Scheduling Behavior Motivation

Scheduling behavior dapat dilakukan dengan renice untuk menaikkan nice value dari sebuah proses. Peningkatan nice value beroperasi terbalik terhadap tingkat prioritas. Proses dengan angka yang lebih tinggi akan bersikap lebih baik atau “nicer” terhadap proses lain sehingga dia akan menurunkan tingkat prioritasnya dibandingkan proses lain.

```
fakhrusy@ubuntudesktop01:~$ yes > /dev/null &
[1] 4043
fakhrusy@ubuntudesktop01:~$ ps -o pid,ni,pri,comm -p 4043,4047
  PID  NI  PRI COMMAND
  4043   0  19 yes
  4047   0  19 yes
fakhrusy@ubuntudesktop01:~$ renice 10 -p 4043
4043 (process ID) old priority 0, new priority 10
fakhrusy@ubuntudesktop01:~$ ps -o pid,ni,pri,%cpu,comm -p 4043,4047
  PID  NI  PRI %CPU COMMAND
  4043  10   9 99.8 yes
  4047   0  19 99.9 yes
```

```
top - 19:33:40 up 30 min, 1 user, load average: 1.96, 1.07, 0.44
Tasks:  2 total,  2 running,  0 sleeping,  0 stopped,  0 zombie
%Cpu(s): 10.9 us, 28.5 sy, 10.8 ni, 49.8 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 7893.8 total, 4487.2 free, 1443.3 used, 2269.8 buff/cache
MiB Swap: 4096.0 total, 4096.0 free,  0.0 used. 6450.5 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
4043	fakhrusy	30	10	16964	2096	1976	R	100.0	0.0	3:51.23	yes
4047	fakhrusy	20	0	16964	2092	1976	R	100.0	0.0	3:44.68	yes

Pin Processes to CPUs and Observe Core-Level Behavior

Taskset digunakan untuk menetapkan hard affinity yang membuat proses tersebut untuk dieksekusi pada CPU tertentu. Akibatnya adalah seluruh eksekusi dari dari instruksi proses tersebut akan terkunci secara eksklusif pada processing unit yang telah ditentukan.

```
4
fakhrusy@ubuntu01:~$ yes > /dev/null &
[1] 4317
fakhrusy@ubuntu01:~$ taskset -p 4317
pid 4317's current affinity mask: f
fakhrusy@ubuntu01:~$ taskset -pc 0 4317
pid 4317's current affinity list: 0-3
pid 4317's new affinity list: 0
fakhrusy@ubuntu01:~$ pidstat -p 4317 1 5
Linux 6.17.0-19-generic (ubuntu01)      04/29/2026      _x86_64_      (4 CPU)

07:44:32 PM  UID      PID    %usr %system %guest  %wait   %CPU   CPU  Command
07:44:33 PM  1000     4317   43.00   57.00   0.00   0.00  100.00    0  yes
07:44:34 PM  1000     4317   43.00   57.00   0.00   0.00  100.00    0  yes
07:44:35 PM  1000     4317   39.00   61.00   0.00   0.00  100.00    0  yes
07:44:36 PM  1000     4317   47.00   54.00   0.00   0.00  101.00    0  yes
07:44:37 PM  1000     4317   43.00   56.00   0.00   0.00   99.00    0  yes
Average:      1000     4317   43.00   57.00   0.00   0.00  100.00    -  yes

fakhrusy@ubuntu01:~$ taskset -pc 0,1 4317
pid 4317's current affinity list: 0
pid 4317's new affinity list: 0,1
fakhrusy@ubuntu01:~$ pidstat -p 4317 1 5
Linux 6.17.0-19-generic (ubuntu01)      04/29/2026      _x86_64_      (4 CPU)

07:45:02 PM  UID      PID    %usr %system %guest  %wait   %CPU   CPU  Command
07:45:03 PM  1000     4317   42.00   58.00   0.00   0.00  100.00    0  yes
07:45:04 PM  1000     4317   43.00   57.00   0.00   0.00  100.00    0  yes
07:45:05 PM  1000     4317   41.00   59.00   0.00   0.00  100.00    0  yes
07:45:06 PM  1000     4317   42.00   58.00   0.00   0.00  100.00    0  yes
07:45:07 PM  1000     4317   46.00   54.00   0.00   0.00  100.00    0  yes
Average:      1000     4317   42.80   57.20   0.00   0.00  100.00    -  yes
```

Explore Real-Time Scheduling Attributes

Proses tersebut berada di bawah policy SCHED_OTHER dengan real time priority 0, scheduling class, dan nilai nice sebesar 0. Ketika mencoba menjalankan proses di bawah policy SCHED_FIFO dengan perintah `chrt -f 10 sleep 30`, sistem mengembalikan error "Operation not permitted". Hasil ini sudah sesuai ekspektasi karena Linux membatasi kemampuan untuk menetapkan real-time scheduling policy hanya kepada pengguna yang ber-privilege.

```
Fakhrusy@ubuntudesktop01:~$ echo $$
3802
Fakhrusy@ubuntudesktop01:~$ chrt -p 3802
pid 3802's current scheduling policy: SCHED_OTHER
pid 3802's current scheduling priority: 0
Fakhrusy@ubuntudesktop01:~$ ps -o pid,cls,rtprio,pri,ni,cmd -p 3802
  PID CLS RTPRIO PRI  NI CMD
 3802 TS      -    19   0 bash
Fakhrusy@ubuntudesktop01:~$ chrt -f 10 sleep 30
chrt: failed to set pid 0's policy: Operation not permitted
```

Follow Program Startup with *strace*, *time*, and Process Mapping

perintah `time ls > /dev/null` menunjukkan bahwa `ls` selesai dieksekusi dalam waktu real `0m0.002s`, dengan user `0m0.000s` dan sys `0m0.002s`. Inspeksi memory map melalui `cat /proc/$!/maps` pada proses `sleep 60` memperlihatkan hasil akhir dari semua operasi `mmap` tersebut dalam bentuk steady-state memory layout.

[illegible]

